SAT Math Mastery · Session 11

Area and Volume

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics area of composite figures · volume of prisms, cylinders, cones, spheres and pyramids · surface area · composite and carved solids · similar figures and solids (scale factors for length, area, volume) · right triangles inside solids · unit conversion · displacement and packing

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos or on a calculator, one expression per line. Where a graph is used, tap a grey dot to read an intersection.

1. HARD · NO CALCULATOR NEEDED

A scale model of a section of a long building is geometrically similar to the actual structure, with a scale factor of 1 to 20 (model to actual). If the volume of the model is 3 cubic meters, what is the volume of the actual segment, in cubic meters?

(A) 8,000 **(B) 24,000 ✓**

(C) 60 (D) 1,200

1. Lengths are 20 times bigger, but volume is a cubic quantity, so the volume is multiplied by $20^3 = 8000$ (not by 20 and not by $20^2 = 400$).

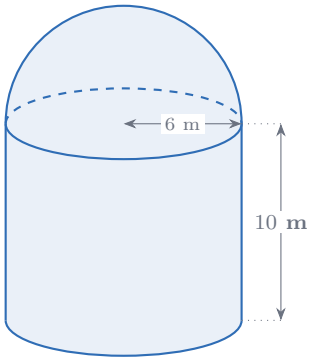
2. Actual volume: $3 \times 8000 = 24000$ cubic meters.

3. That is option (B). The trap answers come from multiplying by 20^2, which gives 1200, or by 20, which gives 60.

Answer: (B)

2. HARD · NO CALCULATOR NEEDED

A pavilion is designed as a cylindrical base topped by a hemispherical dome of the same radius. The cylindrical base has radius 6 meters and height 10 meters, and the dome is a hemisphere with radius 6 meters. What is the total volume of the structure, in cubic meters, in terms of π?



(A) 360π (B) 432π

(C) 504π ✓ (D) 576π

1. Split the building into a cylinder and a hemisphere. Cylinder: $\pi r^2 h = \pi(6)^2(10) = 360\pi$.

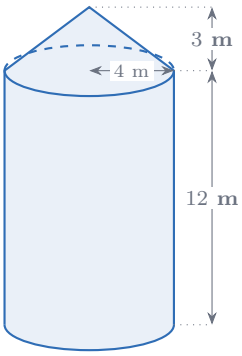
2. Hemisphere (half a sphere): $\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(6)^3 = \frac{2}{3}\pi(216) = 144\pi$.

3. Add: $360\pi + 144\pi = 504\pi$, option (C).

Answer: (C)

3. HARD · NO CALCULATOR NEEDED

A grain silo is designed as a cylindrical base topped by a conical roof of the same radius. The cylindrical base has radius 4 meters and height 12 meters, and the conical roof has radius 4 meters and height 3 meters. What is the total volume of the structure, in cubic meters, in terms of π?



(A) 208π ✓ (B) 240π

(C) 176π (D) 192π

1. Split the silo into a cylinder and a cone. Cylinder: $\pi r^2 h = \pi(4)^2(12) = 192\pi$.

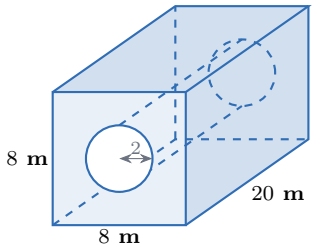
2. Cone: $\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(4)^2(3) = 16\pi$.

3. Add: $192\pi + 16\pi = 208\pi$, option (A).

Answer: (A)

4. HARD · NO CALCULATOR NEEDED

A stone block used in a restoration project is a rectangular prism measuring 8 meters by 8 meters by 20 meters, with a cylindrical channel of radius 2 meters drilled completely through its 20-meter length. What is the volume of the remaining stone, in cubic meters, in terms of π?



(A) $1280 - 160\pi$

(B) $1280 - 20\pi$

(C) $1200 - 80\pi$

(D) $1280 - 80\pi$ ✓

1. Remaining stone = whole prism minus the channel. Prism: $8 \times 8 \times 20 = 1280$.

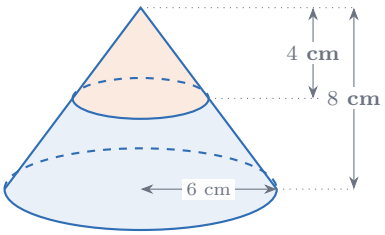
2. The channel is a cylinder of radius 2 and length 20: $\pi r^2 h = \pi(2)^2(20) = 80\pi$.

3. Subtract: $1280 - 80\pi$, option (D).

Answer: (D)

5. HARD · NO CALCULATOR NEEDED

A cone has base radius 6 centimeters and height 8 centimeters, apex pointing up. A plane parallel to the base cuts the cone at a height of 4 centimeters from the apex, removing the smaller cone on top. What is the volume of the removed smaller cone, in cubic centimeters, in terms of π?



(A) 48π

(B) 96π

(C) 12π ✓

(D) 24π

1. The cut is at 4 cm from the apex, half of the full height 8, so the small cone is similar to the big one with scale factor $\frac{1}{2}$. Its radius is 3 cm.

2. Small cone: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(3)^2(4) = 12\pi$.

3. Check with the scale factor: the whole cone is $\frac{1}{3}\pi(6)^2(8) = 96\pi$ and volume scales by $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$, so $96 \div 8 = 12\pi$. Option (C).

Answer: (C)

6. HARD · NO CALCULATOR NEEDED

A cylindrical water tank has a radius of 5 meters and a height of 12 meters. What is the total surface area of the tank, including both circular bases, in square meters, in terms of π?

(A) 150π **(B) 170π ✓**

(C) 220π (D) 120π

1. Total surface area includes both circular bases and the curved side: $SA = 2\pi r^2 + 2\pi rh$.

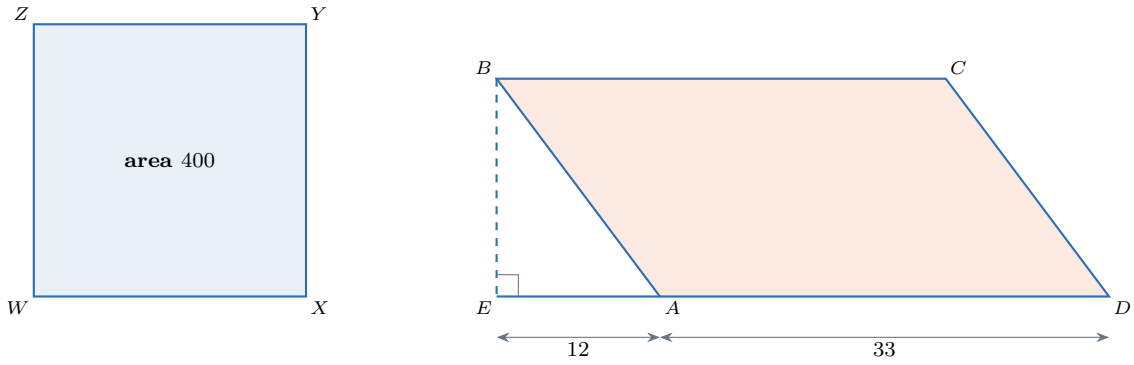
2. Two bases: $2\pi(5)^2 = 50\pi$. Curved side: $2\pi(5)(12) = 120\pi$.

3. Add: $50\pi + 120\pi = 170\pi$, option (B).

Answer: (B)

7. HARD · NO CALCULATOR NEEDED

In the figure, square $WXYZ$ has an area of 400 square units. Side AB of parallelogram $ABCD$ has the same length as a side of the square. Right triangle ABE, formed by dropping an altitude from vertex B to the extension of side AD, has hypotenuse AB and one leg $AE = 12$. If side AD of the parallelogram has length 33, what is the area of parallelogram $ABCD$?



1. The square has area 400, so its side is $\sqrt{400} = 20$. Side AB equals a side of the square, so $AB = 20$.

2. In right triangle ABE the hypotenuse is $AB = 20$ and one leg is $AE = 12$, so $BE = \sqrt{20^2 - 12^2} = \sqrt{256} = 16$. (This is the 3-4-5 triangle scaled by 4: 12, 16, 20.)

3. BE is the height of the parallelogram: it is perpendicular to the line of AD. The area is base times height: $33 \times 16 = 528$.

4. Do not use $AB = 20$ as the height. That is a slanted side, and would give $33 \times 20 = 660$.

Answer: 528

8. HARD · NO CALCULATOR NEEDED

A rectangular vegetable garden originally has length s meters and width $0.4s$ meters. A surrounding path is widened, which decreases the garden's length by $0.1s$ and its width by $0.1s$. What is the percent decrease in the garden's area?

1. Original area: $s \cdot 0.4s = 0.4s^2$.

2. New sides: length $s - 0.1s = 0.9s$ and width $0.4s - 0.1s = 0.3s$. New area: $0.9s \cdot 0.3s = 0.27s^2$.

3. Percent decrease $= \dfrac{\text{old} - \text{new}}{\text{old}} = \dfrac{0.4 - 0.27}{0.4} = 0.325$, which is 32.5%. (Do not subtract the percentages 10% and 25%.)

Answer: 32.5

9. Hard · Calculator

A hemisphere is half of a sphere. If a hemisphere has a radius of 18 inches, which of the following is closest to the volume, in cubic inches, of the hemisphere?

(A) 24,400 (B) 2,400

(C) 6,100 **(D) 12,200 ✓**

1. Hemisphere volume: $V = \frac{2}{3}\pi r^3$ with $r = 18$. First the cube: type `18^3`: it returns 5832.

2. Type `2/3*pi*18^3`: it returns 12214.5122, so the volume is about 12,215 cubic inches.

3. The closest option is 12,200, option (D). The full sphere, $\frac{4}{3}\pi r^3$, would be about 24,429, which is the trap option.

Answer: (D)

10. Hard · Calculator

A conical grain storage silo has a radius of 6 meters and a height of 10 meters. Which of the following is closest to the volume of the silo, in cubic meters? (Use $\pi \approx 3.14$.)

(A) 376.8 ✓ (B) 452.2

(C) 565.2 (D) 282.6

1. Cone volume: $V = \frac{1}{3}\pi r^2 h$ with $\pi \approx 3.14$, $r = 6$, $h = 10$. Type `1/3*3.14*6^2*10`: it returns 376.8.

2. By hand: $6^2 = 36$, and $\frac{1}{3} \cdot 3.14 \cdot 36 \cdot 10 = 376.8$, option (A).

3. Check the traps: forgetting the $\frac{1}{3}$ gives 1130.4, and swapping r and h gives 628.

Answer: (A)

11. Hard · Calculator

A spherical water tank has a radius of 5 meters. Which of the following is closest to the surface area of the tank, in square meters? (Use $\pi \approx 3.14$ and the formula $SA = 4\pi r^2$.)

(A) 261.7 **(B) 314.0 ✓**

(C) 392.5 (D) 235.5

1. Sphere surface area: $SA = 4\pi r^2$ with $\pi \approx 3.14$ and $r = 5$. First $r^2 = 25$.

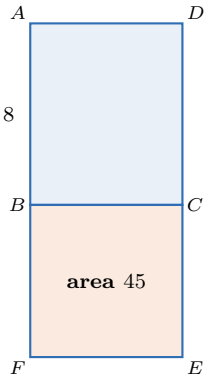
2. Type `4*3.14*5^2`: it returns 314.

3. That is option (B). A wrong formula gives a trap: for example $2\pi r^2$ gives 157.

Answer: (B)

12. HARD · NO CALCULATOR NEEDED

Side AB of rectangle $ABCD$ measures 8 units. Side BC of the rectangle is shared with side BC of square $BCEF$. If the area of square $BCEF$ is 45 square units, what is the area of rectangle $ABCD$?



(A) 80 square units

(B) $12\sqrt{5}$ square units

(C) $24\sqrt{5}$ square units ✓

(D) $45 + 8\sqrt{5}$ square units

1. The square $BCEF$ has area 45, so its side is $BC = \sqrt{45} = \sqrt{9 \cdot 5} = 3\sqrt{5}$ (about 6.708).

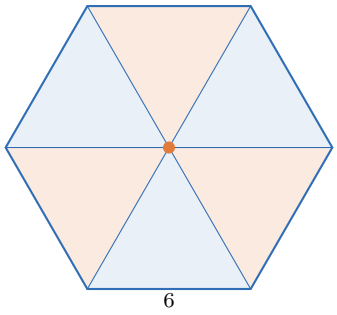
2. The rectangle has sides $AB = 8$ and $BC = 3\sqrt{5}$, so its area is $8 \cdot 3\sqrt{5} = 24\sqrt{5}$ (about 53.666).

3. That is option (C). The option $45 + 8\sqrt{5}$ wrongly adds the two shapes' areas.

Answer: (C)

13. HARD · NO CALCULATOR NEEDED

A hexagon is formed by arranging 6 equilateral triangles, each of which has a vertex at the center of the hexagon. If the length of one side of a triangle is 6, the area of the hexagon is $k\sqrt{3}$ square units. What is the value of k?



1. Each small triangle is equilateral with side 6. Area of an equilateral triangle: $\frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(6)^2 = 9\sqrt{3}$.

2. The hexagon is 6 of these triangles: $6 \times 9\sqrt{3} = 54\sqrt{3}$.

3. The question writes the area as $k\sqrt{3}$, so $k = 54$.

Answer: 54

14. HARD · NO CALCULATOR NEEDED

Two similar rectangles have areas 630 and 2,520. The perimeter of the smaller rectangle is 210, and the perimeter of the larger is n. What is the value of n?

(A) 420 ✓ (B) 2,100

(C) 1,680 (D) 840

1. Area ratio of the similar rectangles: $\dfrac{2520}{630} = 4$.
2. Areas scale by the square of the scale factor, so the scale factor is $\sqrt{4} = 2$.
3. Perimeters scale by the scale factor itself: $n = 2 \times 210 = 420$, option (A). Multiplying by the area ratio 4 would give 840, the trap.

Answer: (A)

15. HARD · NO CALCULATOR NEEDED

A right square pyramid has a base area of 100 and a total surface area of $100 + 100\sqrt{5}$. What is the height of the pyramid?

1. The base is a square of area 100, so its side is 10. The total surface area is base plus lateral, so the four triangular faces have area $100\sqrt{5}$.
2. Each triangle has base 10 and slant height ℓ: $4 \cdot \dfrac{1}{2}(10)\ell = 20\ell = 100\sqrt{5}$, so $\ell = 5\sqrt{5}$ (and $\ell^2 = 125$).
3. Make a right triangle with the height h, half the base 5 and the slant height ℓ: $h^2 + 5^2 = \ell^2$.
4. So $h = \sqrt{125 - 25} = \sqrt{100} = 10$.

Answer: 10

16. HARD · NO CALCULATOR NEEDED

Two similar rectangular crates have surface areas of 144 square feet and 400 square feet. The smaller crate has a volume of 108 cubic feet. What is the volume, in cubic feet, of the larger crate?

(A) 1500 (B) 180

(C) 300 **(D)** 500 ✓

1. Surface areas of similar solids: ratio $\dfrac{400}{144} = \dfrac{25}{9}$, which is the square of the scale factor.
2. So the scale factor is $\sqrt{\dfrac{25}{9}} = \dfrac{5}{3}$, and the volume ratio is its cube: $\left(\dfrac{5}{3}\right)^3 = \dfrac{125}{27}$.
3. Larger volume: $108 \times \dfrac{125}{27} = 4 \times 125 = 500$ cubic feet, option (D).

Answer: (D)

17. HARD · NO CALCULATOR NEEDED

Two similar solid blocks have surface areas of 208 square centimeters and 325 square centimeters. The larger block has a volume of 375 cubic centimeters. What is the volume, in cubic centimeters, of the smaller block?

1. Surface area ratio (small to large): $\dfrac{208}{325} = \dfrac{16}{25}$, the square of the scale factor.
2. Scale factor (small to large): $\sqrt{\dfrac{16}{25}} = \dfrac{4}{5}$. The volume ratio is its cube: $\dfrac{64}{125}$.
3. The smaller block has $375 \times \dfrac{64}{125} = 3 \times 64 = 192$ cubic centimeters.

Answer: 192

18. HARD · DESMOS

A closed right circular cylinder has a volume of 48π cubic centimeters and a total surface area of 56π square centimeters. The height of the cylinder is greater than its diameter. What is the height, in centimeters, of the cylinder?

(A) 3 (B) 4

(C) 12 ✓ (D) 2

1. Volume: $\pi r^2 h = 48\pi$, so $h = \dfrac{48}{r^2}$. Surface area: $2\pi r^2 + 2\pi rh = 56\pi$, so $r^2 + rh = 28$. Substitute h: $r^2 + \dfrac{48}{r} = 28$.

2. In Desmos type `y=x^2+48/x` and `y=28` (letting x stand for r).

3. Tap the intersections with $x > 0$: (2, 28) and (4, 28). (The point with $x = -6$ is not allowed: a radius cannot be negative.)

4. Heights: type `48/2^2`: it returns 12 for $r = 2$, and type `48/4^2`: it returns 3 for $r = 4$. Only $r = 2$ has height greater than the diameter ($12 > 4$). The height is 12, option (C).

Answer: (C)

19. HARD · NO CALCULATOR NEEDED

Sphere A has a great circle with circumference 8π centimeters. The surface area of sphere B is 80π square centimeters greater than the surface area of sphere A. What is the ratio of the volume of sphere B to the volume of sphere A?

(A) $\dfrac{9}{4}$ **(B) $\dfrac{27}{8}$ ✓**

(C) $\dfrac{5}{4}$ (D) $\dfrac{3}{2}$

1. Sphere A: a great circle has circumference $2\pi r = 8\pi$, so $r_A = 4$. Its surface area is $4\pi r^2 = 64\pi$.

2. Sphere B has 80π more: $64\pi + 80\pi = 144\pi$. Then $4\pi r_B^2 = 144\pi$, so $r_B^2 = 36$ and $r_B = 6$.

3. Volumes of spheres scale as the cube of the radius: $\dfrac{V_B}{V_A} = \left(\dfrac{6}{4}\right)^3 = \left(\dfrac{3}{2}\right)^3 = \dfrac{27}{8}$, option (B).

Answer: (B)

20. HARD · NO CALCULATOR NEEDED

A right pyramid has a square base. The diagonal of the base is $10\sqrt{2}$ centimeters and the volume of the pyramid is 400 cubic centimeters. What is the lateral surface area, in square centimeters, of the pyramid (the four triangular faces, not the base)?

(A) 240 (B) 130

(C) 360 **(D) 260 ✓**

1. The base is a square whose diagonal is $10\sqrt{2}$, so its side is $\dfrac{10\sqrt{2}}{\sqrt{2}} = 10$ and its area is 100.

2. Volume $= \dfrac{1}{3}$(base area)h: $400 = \dfrac{1}{3}(100)h$, so $h = \dfrac{3 \cdot 400}{100} = 12$.

3. Slant height: right triangle with the height 12 and half the base 5, so $\ell = \sqrt{12^2 + 5^2} = \sqrt{169} = 13$.

4. Four faces: $4 \cdot \dfrac{1}{2}(10)(13) = 260$ square centimeters, option (D).

Answer: (D)

21. HARD · NO CALCULATOR NEEDED

A metal pipe is a hollow cylinder of length 15 centimeters and outer radius 10 centimeters. The wall has a uniform thickness of t centimeters, where $0 < t < 10$. The volume of metal in the pipe is 765π cubic centimeters. What is the value of t?

1. Metal volume = (outer circle area − inner circle area) times length: $\pi(10^2 - r^2)(15) = 765\pi$, where $r = 10 - t$ is the inner radius.
2. Divide by 15π: $10^2 - r^2 = \dfrac{765}{15} = 51$, so $r^2 = 100 - 51 = 49$.
3. So $r = 7$, and the wall thickness is $t = 10 - r = 10 - 7 = 3$.

Answer: 3

22. HARD · NO CALCULATOR NEEDED

A right pyramid has a rectangular base. Its height is 12 centimeters, and the slant heights of its triangular faces are 13 centimeters for one pair of opposite faces and 20 centimeters for the other pair. What is the total surface area, in square centimeters, of the pyramid, including the base?

(A) 936 ✓ (B) 1160

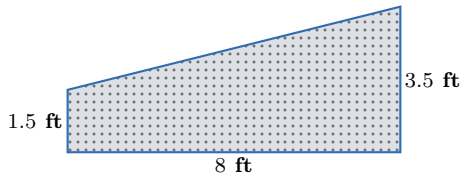
(C) 616 (D) 866

1. Look at the right triangle formed by the height 12, the slant height and the distance from the center of the base to the edge. For slant height 13: distance $= \sqrt{13^2 - 12^2} = 5$. For slant height 20: distance $= \sqrt{20^2 - 12^2} = 16$.
2. So the base measures $2(5) = 10$ by $2(16) = 32$. The faces with slant height 13 sit on the long sides (length 32), the faces with slant height 20 on the short sides (length 10).
3. Face areas: two of $\dfrac{1}{2}(32)(13) = 208$ and two of $\dfrac{1}{2}(10)(20) = 100$, so the lateral area is $416 + 200 = 616$.
4. Add the base $10 \times 32 = 320$: total $616 + 320 = 936$ square centimeters, option (A).

Answer: (A)

23. HARD · CALCULATOR

A raised garden bed is filled with soil. A vertical cross-section of the soil, perpendicular to the 12-foot length of the bed, is a trapezoid: the bed is 8 feet wide, and the soil is 1.5 feet deep along one side and 3.5 feet deep along the opposite side, with the surface sloping uniformly between them. Soil is sold at \$45 per cubic yard and billed for the exact volume (1 yard = 3 feet). What is the cost, in dollars, of the soil needed?



cross-section, perpendicular to the length of the bed

1. The soil's cross-section is a trapezoid with parallel sides 1.5 and 3.5 and width 8. Type `(1.5+3.5)/2*8` : it returns 20 square feet.
2. Volume = cross-section area × length. Type `20*12` : it returns 240 cubic feet.
3. Convert to cubic yards: 1 yard = 3 feet, so 1 cubic yard $= 3^3 = 27$ cubic feet. Type `240/27*45` : it returns 400 dollars.
4. The cost is 400 dollars.

Answer: 400

24. HARD · NO CALCULATOR NEEDED

Cylinder P has radius 2 centimeters and height 6 centimeters. Cylinder Q is similar to cylinder P and has a total surface area of 200π square centimeters. What is the volume, in cubic centimeters, of cylinder Q?

(A) 150π **(B) 375π ✓**

(C) 600π (D) 60π

1. Surface area of P: $2\pi r^2 + 2\pi rh = 2\pi(2)^2 + 2\pi(2)(6) = 8\pi + 24\pi = 32\pi$.
2. Surface areas of similar solids scale by k^2: $k^2 = \dfrac{200}{32} = 6.25$, so the scale factor is $k = 2.5$.
3. Cylinder Q has radius $2 \times 2.5 = 5$ and height $6 \times 2.5 = 15$. Check: $2\pi(5)^2 + 2\pi(5)(15) = 200\pi$.
4. Volume of Q: $\pi r^2 h = \pi(5)^2(15) = 375\pi$, option (B).

Answer: (B)

25. HARD · NO CALCULATOR NEEDED

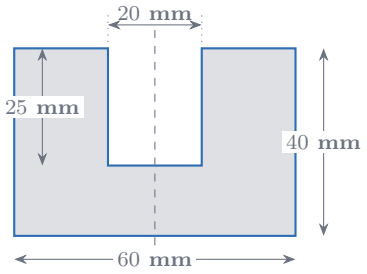
Rectangular box P has dimensions 3 inches by 4 inches by 6 inches. Box Q is similar to box P and has a volume of 1125 cubic inches. What is the total surface area, in square inches, of box Q?

1. Box P: volume $3 \cdot 4 \cdot 6 = 72$ and surface area $2(12 + 18 + 24) = 108$.
2. Volume ratio $\dfrac{1125}{72} = 15.625$, which is the cube of the scale factor k. So $k = \sqrt[3]{15.625} = 2.5$.
3. Surface areas scale by $k^2 = 6.25$: $108 \times 6.25 = 675$ square inches.
4. Check: box Q is $7.5 \times 10 \times 15$ and $2(75 + 112.5 + 150) = 675$.

Answer: 675

26. HARD · CALCULATOR

A brass bushing is a solid right circular cylinder with diameter 60 millimeters and height 40 millimeters. A cylindrical hole of diameter 20 millimeters is drilled into one flat face, along the axis, to a depth of 25 millimeters. The density of brass is 7.8 grams per cubic centimeter. Which of the following is closest to the mass of the bushing, in grams? (1 cm = 10 mm)



cross-section through the axis

(A) 82,090 (B) 784

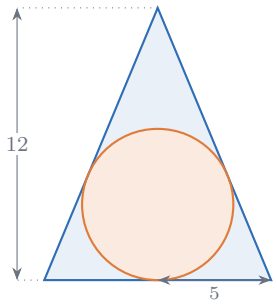
(C) 821 ✓ (D) 882

1. Convert to centimeters first (10 mm = 1 cm): the cylinder has radius 3 cm and height 4 cm, and the hole has radius 1 cm and depth 2.5 cm.
2. Whole cylinder: Type `pi*3^2*4`: it returns 113.0973 (cm^3). Hole: type `pi*1^2*2.5`: it returns 7.854 (cm^3).
3. Metal left: Type `pi*3^2*4-pi*1^2*2.5`: it returns 105.2434 (cm^3).
4. Mass = volume × density. Type `(pi*3^2*4-pi*1^2*2.5)*7.8`: it returns 820.8982, so about 821 grams, option (C). Ignoring the hole gives 882, the trap.

Answer: (C)

27. HARD · NO CALCULATOR NEEDED

A sphere is inscribed in a right circular cone whose base has radius 5 and whose height is 12, so that the sphere is tangent to the base and to the lateral surface of the cone. What fraction of the volume of the cone is occupied by the sphere? (Enter your answer as a fraction.)



1. Cut the cone and sphere through the axis: you see an isosceles triangle (base $2 \times 5 = 10$, height 12) with the sphere as its inscribed circle. The slanted side is $\sqrt{5^2 + 12^2} = 13$.
2. The inscribed circle's radius is $\dfrac{\text{area}}{\text{semiperimeter}} = \dfrac{\frac{1}{2}(10)(12)}{(13+13+10)/2} = \dfrac{60}{18} = \dfrac{10}{3}$.
3. Sphere volume: $\dfrac{4}{3}\pi\left(\dfrac{10}{3}\right)^3 = \dfrac{4000\pi}{81}$. Cone volume: $\dfrac{1}{3}\pi(5)^2(12) = 100\pi$.
4. Fraction occupied: $\dfrac{4000/81}{100} = \dfrac{40}{81}$.

Answer: $\dfrac{40}{81}$

28. HARD · NO CALCULATOR NEEDED

A museum has a small solid model of a sculpture and a full-size copy; the two are similar solids. The sealant needed to coat the whole surface is proportional to surface area. The model needs 1.6 liters of sealant and the full-size copy needs 14.4 liters. The model has a volume of 2.5 cubic feet. What is the volume, in cubic feet, of the full-size copy?

(A) 67.5 ✓ (B) 202.5

(C) 7.5 (D) 22.5

1. Sealant is proportional to surface area, so the surface area ratio is $\dfrac{14.4}{1.6} = 9$.
2. Surface area scales by k^2, so the scale factor is $k = \sqrt{9} = 3$.
3. Volume scales by $k^3 = 3^3 = 27$: $2.5 \times 27 = 67.5$ cubic feet, option (A).

Answer: (A)

29. HARD · CALCULATOR

Two aquariums are similar in shape. The smaller aquarium holds 250 liters and has 400 square decimeters of glass in total. The larger aquarium holds 432 liters. How many square decimeters of glass does the larger aquarium have in total? (Note that 1 liter = 1 cubic decimeter.)

1. Volume ratio of the aquariums: $\dfrac{432}{250} = 1.728$. Volumes scale by k^3, so find k: Type `(432/250)^(1/3)` : it returns 1.2.
2. Glass area scales by k^2. Type `400*1.2^2` : it returns 576 square decimeters.
3. The larger aquarium has 576 square decimeters of glass. (Using k^3 instead would give 691.2.)

Answer: 576

30. HARD · NO CALCULATOR NEEDED

A roof ornament is a solid pyramid whose base is a square of area 100 square inches. The apex is directly above the center of the base, and each of the four edges from the apex to a corner of the base has length 13 inches. What is the combined area, in square inches, of the four triangular lateral faces?

(A) 260 (B) 120

(C) 340 **(D) 240 ✓**

1. The base has area 100, so its side is $\sqrt{100} = 10$. Each triangular face has base 10 and two equal sides of length 13 (the edges from the apex).
2. Draw the height of one face, from the apex edge down to the middle of the base. It splits the base in halves of 5, so $h_{\text{face}} = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$.
3. One face: $\frac{1}{2}(10)(12) = 60$. Four faces: $4 \times 60 = 240$ square inches, option (D).
4. The trap: treating 13 as the face height gives $4 \cdot \frac{1}{2}(10)(13) = 260$.

Answer: (D)

31. HARD · NO CALCULATOR NEEDED

Two similar rectangular prisms have volumes in the ratio $8 : 27$. The smaller prism has a surface area of 40. What is the surface area of the larger prism?

1. Volumes of similar solids are in the ratio $8 : 27 = 2^3 : 3^3$, so the lengths are in the ratio $2 : 3$, that is the scale factor is $\frac{3}{2}$.
2. Surface areas scale by the square: $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$.
3. Larger surface area: $40 \times \frac{9}{4} = 90$.

Answer: 90

32. HARD · CALCULATOR

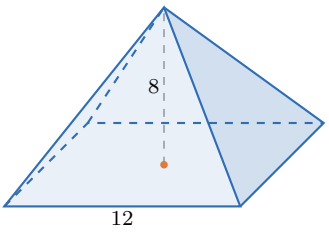
A rectangular tank has a base that is 40 centimeters by 30 centimeters. A solid cube with edge 12 centimeters is placed in the tank and sinks to the bottom, fully under water. By how many centimeters does the water level rise?

1. The water level rises because the cube pushes aside its own volume of water. Type `12^3`: it returns 1728 cubic centimeters.
2. That extra volume spreads over the base of the tank. Type `40*30`: it returns 1200 square centimeters.
3. Rise $= \frac{\text{volume}}{\text{base area}}$: Type `1728/1200`: it returns 1.44 centimeters.

Answer: 1.44

33. HARD · NO CALCULATOR NEEDED

A square pyramid has a base with side length 12 and a height of 8. What is the lateral surface area of the pyramid (the four triangular faces, excluding the base)?



1. Each triangular face has base 12. Its height (the slant height) comes from a right triangle with the pyramid's height 8 and half the base 6: $\sqrt{8^2+6^2}=\sqrt{100}=10$.
2. One face: $\frac{1}{2}(12)(10)=60$.
3. Four faces: $4\times 60=240$. Do not use the height 8 as the face height: that would give $4\cdot\frac{1}{2}(12)(8)=192$.

Answer: 240

34. HARD · CALCULATOR

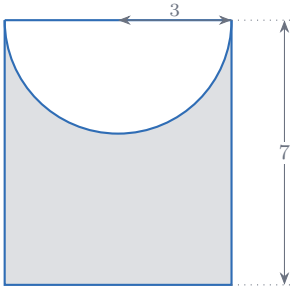
The radius of a sphere is increased by 20%. By what percent does the volume of the sphere increase? (Enter as a decimal.)

1. Increasing the radius by 20% multiplies it by 1.2. Volume depends on r^3, so the volume is multiplied by 1.2^3: Type `1.2^3`: it returns 1.728.
2. A multiplier of 1.728 means an increase of 72.8%: Type `(1.2^3-1)*100`: it returns 72.8.
3. Enter 72.8. (The trap $3\times 20=60$ forgets that the change compounds.)

Answer: 72.8

35. HARD · NO CALCULATOR NEEDED

A solid is formed by carving a hemispherical bowl of radius 3 out of the top of a right circular cylinder of radius 3 and height 7. The volume of the solid is $k\pi$. What is the value of k?



cross-section through the axis

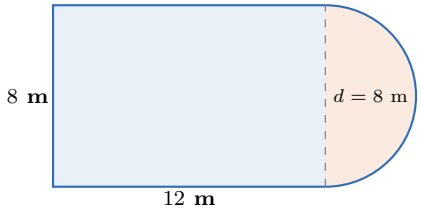
1. The solid is the cylinder with the hemispherical bowl taken away. Cylinder: $\pi r^2 h=\pi(3)^2(7)=63\pi$.
2. Hemisphere (the bowl): $\frac{2}{3}\pi r^3=\frac{2}{3}\pi(3)^3=18\pi$.
3. Subtract: $63\pi-18\pi=45\pi$, so $k=45$.

Answer: 45

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. Medium · No calculator needed

A school courtyard is shaped like a 12-meter by 8-meter rectangle with a semicircular reading nook attached to one of the 8-meter sides, as shown. The semicircle has a diameter of 8 meters. What is the total area of the courtyard, in square meters?



(A) 104π

(B) $96 + 4\pi$

(C) $96 + 8\pi$ ✓

(D) $96 + 16\pi$

1. The courtyard is a rectangle plus a half circle. The rectangle has area $12 \times 8 = 96$.

2. The half circle has diameter 8, so its radius is 4 (not 8). Its area is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(4)^2 = 8\pi$.

3. Add the two pieces: $96 + 8\pi$, which is option (C). Using the diameter as the radius would give $\frac{1}{2}\pi(8)^2 = 32\pi$, the trap.

Answer: (C)

37. Medium · No calculator needed

A cylindrical water tank supplying a public fountain has a radius of 15 meters and a volume of 4500π cubic meters. What is the height of the tank, in meters?

(A) 15

(B) 20 ✓

(C) 30

(D) 10

1. Volume of a cylinder: $V = \pi r^2 h$. Put in $r = 15$ and $V = 4500\pi$: $4500\pi = \pi(15)^2 h = 225\pi h$.

2. Divide both sides by 225π: $h = \frac{4500}{225} = 20$.

3. The height is 20 meters, option (B).

Answer: (B)

38. Medium · No calculator needed

A rectangular storage box has length 8 cm, width 5 cm, and height 4 cm. What is the total surface area of the box, in square centimeters?

(A) 92

(B) 144

(C) 160

(D) 184 ✓

1. A box has 6 faces in 3 pairs: length by width, length by height, width by height.

2. The three face areas: $8 \times 5 = 40$, $8 \times 4 = 32$, $5 \times 4 = 20$. Their sum is 92.

3. Each face appears twice: $2 \times 92 = 184$ square centimeters, option (D).

Answer: (D)

39. Medium · No calculator needed

A cube has a total surface area of 150 square centimeters. What is the volume of the cube, in cubic centimeters?

(A) 125 ✓ (B) 150

(C) 25 (D) 75

1. A cube has 6 equal square faces, so $6s^2 = 150$ and $s^2 = \dfrac{150}{6} = 25$.

2. Then $s = \sqrt{25} = 5$ centimeters.

3. Volume: $s^3 = 5^3 = 125$, option (A).

Answer: (A)

40. Medium · No calculator needed

A rectangular water storage tank at a facility has a base measuring 8 feet by 5 feet. The tank is filled with water to a depth of 4.5 feet. Using a water density of 62.4 pounds per cubic foot, what is the weight of the water in the tank, in pounds?

1. Volume of water = base area times depth. Type `8*5*4.5` : it returns 180 cubic feet.

2. Weight = volume times density. Type `180*62.4` : it returns 11232 pounds.

3. Enter 11232 (no comma) as the answer.

Answer: 11232

41. Medium · No calculator needed

A cylindrical water tank has a base circumference of 44π meters and a height of 15 meters. The volume of the tank is $k\pi$ cubic meters. What is the value of k?

1. The circumference of the base is $2\pi r = 44\pi$, so $r = \dfrac{44\pi}{2\pi} = 22$ meters.

2. Volume of a cylinder: $V = \pi r^2 h = \pi(22)^2(15)$. Type `22^2*15` : it returns 7260, so $V = 7260\pi$.

3. The question writes the volume as $k\pi$, so $k = 7260$.

Answer: 7260

42. Medium · No calculator needed

A design team is producing rectangular flag replica banners of two sizes. Rectangle A is similar to rectangle B, and the side lengths of rectangle A are 3 times the corresponding side lengths of rectangle B. If the area of rectangle B is 24 square centimeters, what is the area, in square centimeters, of rectangle A?

1. Similar rectangles: lengths scale by 3, so areas scale by $3^2 = 9$.

2. Area of $A = 24 \times 9 = 216$ square centimeters.

3. The trap answer $24 \times 3 = 72$ forgets that area is two-dimensional.

Answer: 216

43. Medium · No calculator needed

For a school engineering fair project, a team builds a right circular cone-shaped sand mound model with a volume of 48π cubic centimeters and a height of 9 centimeters. What is the radius, in centimeters, of the base of the cone?

1. Cone: $V = \dfrac{1}{3}\pi r^2 h$. Put in $V = 48\pi$ and $h = 9$: $48\pi = \dfrac{1}{3}\pi r^2(9) = 3\pi r^2$.

2. Divide by 3π: $r^2 = \dfrac{48}{3} = 16$.

3. So $r = \sqrt{16} = 4$ centimeters.

Answer: 4

44. Medium · Calculator

A cylindrical water tank has a radius of 3 meters and a height of 8 meters. Which of the following is closest to the volume of the tank, in cubic meters? (Use $\pi \approx 3.14$.)

(A) 188.4

(B) 226.1 ✓

(C) 254.3

(D) 150.7

1. Cylinder: $V = \pi r^2 h$ with $\pi \approx 3.14$, $r = 3$, $h = 8$. Type `3.14*3^2*8`: it returns 226.08.

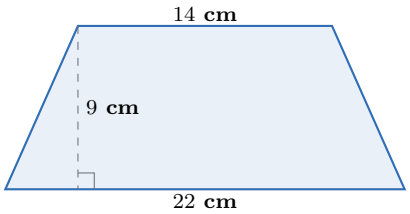
2. Round to one decimal place: 226.1, option (B).

3. Check the traps: forgetting to square r gives 75.4, and using the diameter 6 as the radius gives 904.3.

Answer: (B)

45. Medium · No calculator needed

A trapezoid has parallel sides of length 14 cm and 22 cm, and a height of 9 cm. What is the area of the trapezoid, in square centimeters?



(A) 144

(B) 153

(C) 162 ✓

(D) 171

1. Trapezoid area: $\frac{1}{2}(b_1 + b_2)h$, where b_1 and b_2 are the parallel sides and h is the height (the perpendicular distance).

2. Add the parallel sides: $14 + 22 = 36$.

3. Area $= \frac{1}{2}(36)(9) = 162$ square centimeters, option (C).

Answer: (C)

46. Medium · No calculator needed

For a particular square, the perimeter is numerically equal to the area. What is the perimeter of the square?

1. Let the side be s. Perimeter $= 4s$ and area $= s^2$. Set them equal: $s^2 = 4s$.

2. Rearrange: $s^2 - 4s = 0$, so $s(s - 4) = 0$. The side cannot be 0, so $s = 4$.

3. The perimeter is $4 \times 4 = 16$.

Answer: 16

47. Medium · No calculator needed

A circular coaster has a diameter of 10 cm and a height (thickness) of 3 mm. If 8 coasters are stacked on top of each other, what is the volume of the stack?

(A) 60π **cm**^{3} ✓ (B) 75π cm^3

(C) 120π cm^3 (D) 150π cm^3

1. Diameter 10 cm means radius 5 cm. The thickness is 3 mm, which is 0.3 cm (convert before multiplying).

2. One coaster: $\pi r^2 h = \pi(5)^2(0.3) = 7.5\pi$ cubic centimeters.

3. Eight coasters: $8 \times 7.5\pi = 60\pi$ cubic centimeters, option (A).

Answer: (A)

48. Medium · No calculator needed

If the surface area of a cube is 96 square centimeters, what is the volume of the cube, in cubic centimeters?

1. A cube has 6 equal faces: $6s^2 = 96$, so $s^2 = \dfrac{96}{6} = 16$.

2. Then $s = \sqrt{16} = 4$ centimeters.

3. Volume: $s^3 = 4^3 = 64$ cubic centimeters.

Answer: 64

49. Medium · No calculator needed

Two similar rectangular prisms have volumes 972 and 36. An edge of the larger prism has length 12. What is the length of the corresponding edge of the smaller prism?

1. Volumes of similar solids: the ratio is the scale factor cubed. $\dfrac{972}{36} = 27$.

2. The scale factor is $\sqrt[3]{27} = 3$ (because $3^3 = 27$).

3. The larger edge is 12, so the smaller edge is $\dfrac{12}{3} = 4$.

Answer: 4

50. Medium · No calculator needed

The edges of cubes A and B are in the ratio $\dfrac{3}{88}$ (edge of A to edge of B). What is the ratio of the surface area of B to the surface area of A?

(A) $\dfrac{9}{7{,}744}$ (B) $\dfrac{3}{88}$

(C) $\dfrac{88}{3}$ **(D)** $\dfrac{7{,}744}{9}$ ✓

1. The question asks for B to A, so flip the given ratio: edge of B to edge of A is $\dfrac{88}{3}$.

2. Surface areas scale by the square of the edge ratio: $\left(\dfrac{88}{3}\right)^2 = \dfrac{88^2}{3^2} = \dfrac{7744}{9}$.

3. That is option (D). Options with $\dfrac{3}{88}$ or $\dfrac{88}{3}$ forget to square, and $\dfrac{9}{7744}$ has B and A the wrong way round.

Answer: (D)

51. Medium · No calculator needed

A right circular cylinder has volume 90π and height 10. The lateral surface area (excluding the bases) of the cylinder is $k\pi$. What is the value of k?

1. Volume: $\pi r^2(10) = 90\pi$, so $r^2 = \dfrac{90}{10} = 9$ and $r = 3$.

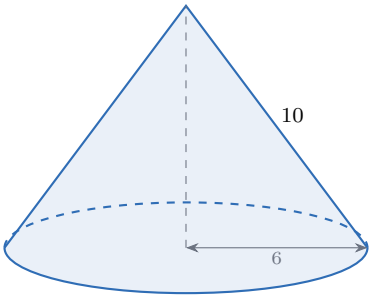
2. Lateral surface (no bases): $2\pi rh = 2\pi(3)(10) = 60\pi$.

3. So $k = 60$.

Answer: 60

52. Medium · No calculator needed

A right circular cone has base radius 6 and slant height 10. The volume of the cone is $k\pi$. What is the value of k?



1. The radius, height and slant height form a right triangle: $h = \sqrt{10^2 - 6^2} = \sqrt{64} = 8$.

2. Volume: $\dfrac{1}{3}\pi r^2 h = \dfrac{1}{3}\pi(36)(8) = 96\pi$.

3. So $k = 96$.

Answer: 96

53. Medium · No calculator needed

A sphere has surface area 144π. The volume of the sphere is $k\pi$. What is the value of k?

1. Surface area: $4\pi r^2 = 144\pi$, so $r^2 = \dfrac{144}{4} = 36$ and $r = 6$.

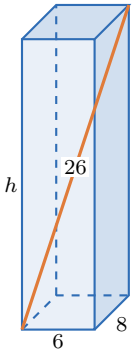
2. Volume: $\dfrac{4}{3}\pi r^3 = \dfrac{4}{3}\pi(6)^3 = \dfrac{4}{3}\pi(216) = 288\pi$.

3. So $k = 288$.

Answer: 288

54. Medium · No calculator needed

A rectangular box has a base that measures 6 by 8. The distance from one corner of the box to the opposite corner (the space diagonal) is 26. What is the height of the box?



1. The space diagonal d of a box satisfies $d^2 = \text{length}^2 + \text{width}^2 + h^2$, so $6^2 + 8^2 + h^2 = 26^2$.

2. Compute: $100 + h^2 = 676$, so $h^2 = 576$.

3. So $h = \sqrt{576} = 24$.

Answer: 24

55. Medium · No calculator needed

The space diagonal of a cube (the segment joining opposite vertices) has length $9\sqrt{3}$. What is the volume of the cube?

1. The space diagonal of a cube with edge e is $e\sqrt{3}$. So $e\sqrt{3} = 9\sqrt{3}$ and $e = 9$.

2. Volume: $e^3 = 9^3 = 729$.

3. Check the trap: the face area $9^2 = 81$ is not the volume.

Answer: 729

56. Medium · No calculator needed

A box has interior dimensions 10 cm by 12 cm by 16 cm. Cubes with edge length 2 cm are packed into the box with no gaps. How many cubes fit?

1. Count how many cubes fit along each edge: $\frac{10}{2} = 5$, $\frac{12}{2} = 6$, $\frac{16}{2} = 8$.

2. Cubes fit in a $5 \times 6 \times 8$ grid, so the total is $5 \times 6 \times 8 = 240$.

3. Check by volumes: $\frac{10 \cdot 12 \cdot 16}{2^3} = \frac{1920}{8} = 240$.

Answer: 240

57. Medium · No calculator needed

A cone and a cylinder each have radius 5 and height 12. The cone is filled with water and the water is poured into the cylinder, repeatedly, until the cylinder is full. How many full cones of water does it take to fill the cylinder?

1. Cylinder volume: $\pi r^2 h = \pi(5)^2(12) = 300\pi$.

2. Cone volume (one third of the cylinder with the same base and height): $\frac{1}{3}\pi(5)^2(12) = 100\pi$.

3. Number of cones: $\frac{300\pi}{100\pi} = 3$.

Answer: 3